

Supplementary Data. The regulatory network controlling dendritic cells differentiation

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Abbreviations: CDP, common dendritic progenitos. pre-cDC, pre- classical Dendritic Cell. pre-cDC1, pre- classical Dendritic Cell type 1. CD4+ cDCs and CD8+ cDCs, CD4+, and CD8+ classical dendritic cells respectively. pDC plasmacytoid dendritic cell. ND, Not defined (numeration stats again in every table).

Supplementary Table 1. BCL6-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP-like	0	1	0	0	1	1	0	1	1	0	1	1	0	1	0	0	1	1	1	1	0
pre-cDC-like	0	1	1	0	1	1	0	1	1	0	1	1	0	1	0	0	1	1	1	1	0
pre-cDC1-like	0	1	0	0	1	1	0	1	1	1	0	1	0	1	1	0	1	1	1	1	1
cDC CD8+-like	0	1	1	0	1	1	0	1	1	1	0	1	0	1	1	0	1	1	1	1	1

Supplementary Table 2. BCL6-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	0	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	0	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	0	1	0	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 3. BATF3-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	1	0	0	0	1	1	0	1	0	0	1	0	0	1	0	0	1	1	1	1	0
ND2	1	0	1	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
pre-cDC1-like	1	0	0	0	1	1	0	1	1	1	0	0	0	1	0	0	1	1	1	1	1
cDC CD8+-like	1	0	1	0	1	1	0	1	1	1	0	0	0	1	0	0	1	1	1	1	1

Supplementary Table 4. BATF3-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	0	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	0	1	0	1	1	0	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	0	1	0	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 5. CD8-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	1	1	1	0	1	1	0	1	1	0	0	0	1	1	0	0	1	1	1	1	1
ND2	1	1	1	0	1	1	0	1	0	0	1	0	0	1	0	0	1	1	1	1	0

Supplementary Table 6. CD8-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	1	1	0	0	1	1	0	1	1	0	0	0	1	1	0	0	1	1	1	1	1
ND2	1	1	0	0	1	1	0	1	0	0	1	0	0	1	0	0	1	1	1	1	0

Supplementary Table 7. E2-2-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP-like	1	1	0	1	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
ND1	1	1	1	1	1	1	0	1	0	0	1	0	0	1	0	0	1	1	1	1	0
ND2	1	1	0	1	1	1	0	1	1	0	0	0	1	1	0	0	1	1	1	1	1
ND3	1	1	1	1	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1

Supplementary Table 8. E2-2-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	0	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	0	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 9. FLT3-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	0	1	0	0	0	1	0	1	1	0	0	0	0	1	0	0	1	1	0	1	0
ND2	0	1	1	0	0	1	0	1	1	0	0	0	0	1	0	0	1	1	0	1	0
ND3	1	1	0	0	0	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
ND4	1	1	1	0	0	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
ND5	1	1	0	0	0	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1
ND6	1	1	1	0	0	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1

Supplementary Table 10. FLT3-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 11. FLT3L-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 12. FLT3L-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 13. GATA1-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP-like	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	1	1	1	1	1	0
pre-cDC-like	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	1	1	1	1	1	0
ND1	1	1	0	0	1	1	1	1	1	1	0	0	0	1	0	1	1	1	1	1	1
ND2	1	1	1	0	1	1	1	1	1	1	0	0	0	1	0	1	1	1	1	1	1

Supplementary Table 14. GATA1-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 15. GM-CSF-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	1	0	0	1	1	1	0	0	0	1	0	1	0	1	0	0	1	1	1	0	1
ND2	1	0	1	1	1	1	0	0	0	1	0	1	0	1	0	0	1	1	1	0	1
ND3	1	0	0	1	1	1	0	0	0	1	0	0	1	0	0	1	1	1	1	0	1
ND4	1	0	1	1	1	1	0	0	0	1	0	0	1	0	0	1	1	1	1	0	1
ND5	1	1	0	0	1	1	0	0	1	1	0	1	1	1	0	0	1	1	1	1	1
ND6	1	1	1	0	1	1	0	0	1	1	0	1	1	1	0	0	1	1	1	1	1
ND7	1	1	0	0	1	1	0	0	1	0	1	0	0	1	0	0	1	1	1	1	1
ND8	1	1	1	0	1	1	0	0	1	0	1	0	0	1	0	0	1	1	1	1	1

Supplementary Table 16. GM-CSF-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 17. ID2-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
pDC-like	1	0	0	1	1	1	0	1	0	0	1	0	0	1	0	0	1	1	1	1	0
ND1	1	0	1	1	1	1	0	1	0	0	1	0	0	1	0	0	1	1	1	1	0
ND2	1	0	0	1	1	1	0	1	0	1	0	0	0	1	0	0	1	1	1	1	1
ND3	1	0	1	1	1	1	0	1	0	1	0	0	0	1	0	0	1	1	1	1	1

Supplementary Table 18. ID2-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 19. IFN α/β -gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP-like	1	1	0	0	1	1	0	1	1	1	1	0	1	1	0	0	1	1	1	1	1
pre-cDC-like	1	1	1	0	1	1	0	1	1	1	1	0	1	1	0	0	1	1	1	1	1
pre-cDC1-like	1	1	0	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1
cDC CD8+-like	1	1	1	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1

Supplementary Table 20. IFN α/β -loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP-like	1	1	0	0	1	1	0	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC-like	1	1	1	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
ND1	1	1	0	0	1	1	0	1	1	0	0	0	0	1	0	0	1	1	1	1	0
ND2	1	1	1	0	1	1	0	1	1	0	0	0	0	1	0	0	1	1	1	1	0

Supplementary Table 21. IL-10-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	1	1	0	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
ND2	1	1	1	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0

Supplementary Table 22. IL-10-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	1	1	0	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1
ND2	1	1	1	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1

Supplementary Table 23. IL-12-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	1	1	0	0	1	1	0	1	1	0	1	1	0	1	0	0	1	1	1	1	0
ND2	1	1	1	0	1	1	0	1	1	0	1	1	0	1	0	0	1	1	1	1	0
pre-cDC1-like	1	1	0	0	1	1	0	1	1	1	0	1	1	1	0	0	1	1	1	1	1
cDC CD8+-like	1	1	1	0	1	1	0	1	1	1	0	1	1	1	0	0	1	1	1	1	1

Supplementary Table 24. IL-12-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP-like	1	1	0	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
pre-cDC-like	1	1	1	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
ND1	1	1	0	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1
ND2	1	1	1	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1

Supplementary Table 25. IRF4-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
pre-cDC-like	1	1	0	0	1	1	0	1	1	0	1	0	1	1	0	0	1	1	1	1	0
ND1	1	1	1	0	1	1	0	1	1	0	1	0	1	1	0	0	1	1	1	1	0
ND2	1	1	0	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1
ND3	1	1	1	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1

Supplementary Table 26. IRF4-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	1	1	0	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
ND2	1	1	1	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
ND3	1	1	0	0	1	1	0	1	1	1	0	0	0	1	0	0	1	1	1	1	1
ND4	1	1	1	0	1	1	0	1	1	1	0	0	0	1	0	0	1	1	1	1	1

Supplementary Table 27. IRF8-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
cDC CD4+-like	1	1	0	0	1	1	0	1	1	0	1	0	1	0	0	0	1	1	1	1	0
ND1	1	1	1	0	1	1	0	1	1	0	1	0	1	0	0	0	1	1	1	1	0
ND2	1	1	0	0	1	1	0	1	1	1	0	0	1	0	0	0	1	1	1	1	1
ND3	1	1	1	0	1	1	0	1	1	1	0	0	1	0	0	0	1	1	1	1	1

Supplementary Table 28. IRF8-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 29. NF- κ B-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	1	1	0	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1	1	1	0
ND2	1	1	1	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1	1	1	0
pre-cDC1-like	1	1	0	0	1	1	0	1	1	1	0	1	1	1	1	0	1	1	1	1	1
cDC CD8+-like	1	1	1	0	1	1	0	1	1	1	0	1	1	1	1	0	1	1	1	1	1

Supplementary Table 30. NF- κ B-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 31. NFIL3-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP-like	1	1	0	0	1	1	0	1	1	0	1	0	0	1	0	1	1	1	1	1	0
pre-cDC-like	1	1	1	0	1	1	0	1	1	0	1	0	0	1	0	1	1	1	1	1	0
pre-cDC1-like	1	1	0	0	1	1	0	1	1	1	0	0	1	1	0	1	1	1	1	1	1
cDC CD8+-like	1	1	1	0	1	1	0	1	1	1	0	0	1	1	0	1	1	1	1	1	1

Supplementary Table 32. NFIL3-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 33. PU.1-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	0	1	0	0	0	1	0	1	1	0	0	0	0	1	0	0	0	1	0	1	0
ND2	0	1	1	0	0	1	0	1	1	0	0	0	0	1	0	0	1	0	0	1	0
ND3	0	1	0	0	0	1	1	1	1	0	0	0	0	1	0	0	0	1	0	1	0
ND4	0	1	1	0	0	1	1	1	1	0	0	0	0	1	0	0	0	1	0	0	0
ND5	1	1	0	0	0	1	0	1	1	0	1	0	0	1	0	0	0	1	1	1	0
ND6	1	1	1	0	0	1	0	1	1	0	1	0	0	1	0	0	0	1	1	1	0
ND7	1	1	0	0	0	1	1	1	1	0	1	0	0	1	0	1	0	1	1	1	0
ND8	1	1	1	0	0	1	1	1	1	0	1	0	0	1	0	1	0	1	1	1	0
ND9	1	1	0	0	0	1	0	1	1	1	0	0	1	1	0	0	0	1	1	1	1
ND10	1	1	1	0	0	1	0	1	1	1	0	0	1	1	0	0	0	1	1	1	1
ND11	1	1	0	0	0	1	1	1	1	1	0	0	1	1	0	1	0	1	1	1	1
ND12	1	1	1	0	0	1	1	1	1	1	0	0	1	1	0	1	0	1	1	1	1

Supplementary Table 34. PU.1-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 35. RNUX1-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP-like	1	1	0	0	1	1	0	1	1	0	1	0	0	1	0	0	1	0	1	1	0
pre-cDC-like	1	1	1	0	1	1	0	1	1	0	1	0	1	1	0	0	1	0	1	1	0
pre-cDC1-like	1	1	0	0	1	1	0	1	1	1	0	0	1	1	0	0	1	0	1	1	1
cDC CD8+-like	1	1	1	0	1	1	0	1	1	1	0	0	1	1	0	0	1	0	1	1	1

Supplementary Table 36. RUNX1-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 37. STAT3-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	0	1	0	0	1	1	0	1	1	0	0	0	1	1	0	0	1	1	0	1	0
ND2	0	1	1	0	1	1	0	1	1	0	0	0	1	1	0	0	1	1	0	1	0
ND3	0	1	0	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	0	1	0
ND4	0	1	1	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	0	1	0

Supplementary Table 38. STAT3-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 39. STAT5-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
pDC-like	1	0	0	1	1	1	0	1	0	0	1	0	0	1	0	0	1	0	1	0	0
ND1	1	0	1	1	1	1	0	1	0	0	1	0	0	1	0	0	1	1	1	0	0
ND2	1	0	0	1	1	1	0	1	0	1	0	0	0	1	0	0	1	0	1	0	1
ND3	1	0	1	1	1	1	0	1	0	1	0	0	0	1	0	0	1	1	1	0	1

Supplementary Table 40. STAT5-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP	1	1	0	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC	1	1	1	0	1	1	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0
pre-cDC1	1	1	0	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1
cDC CD8+	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	0	1	1	1	1	1

Supplementary Table 41. TLR3-loss mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	1	1	0	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
ND2	1	1	1	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	0
ND3	1	1	0	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	0
ND4	1	1	1	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	0

Supplementary Table 42 TLR3-gain mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
CDP-like	1	1	0	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	1
pre-cDC-like	1	1	1	0	1	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1	1
pre-cDC1-like	1	1	0	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1
cDC CD8 ⁺ -like	1	1	1	0	1	1	0	1	1	1	0	0	1	1	0	0	1	1	1	1	1

Supplementary Table 43. FLT3L-loss/GM-CSF-loss double mutant model.

	BCL6	BATF3	CD8	E2-2	FLT3	FLT3L	GATA1	GM-CSF	ID2	IFN α/β	IL-10	IL-12	IRF4	IRF8	NF- κ B	NFIL3	PU.1	RUNX1	STAT3	STAT5	TLR3
ND1	0	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0
ND2	0	0	0	1	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0
ND3	0	0	1	1	0	0	0	0	0	0	0	1	1	0	0	0	0	1	0	0	0
ND4	0	0	1	1	0	0	0	0	0	0	0	1	1	1	0	0	0	1	0	0	0
ND5	1	0	0	1	1	0	0	0	0	1	0	1	0	1	0	0	1	1	1	0	1
ND6	1	0	1	1	1	0	0	0	0	1	0	1	0	1	0	0	1	1	1	0	1
ND7	0	0	0	1	0	0	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0
ND8	0	0	0	1	0	0	1	0	0	0	0	1	1	1	0	0	0	0	0	0	0
ND9	0	0	1	1	0	0	1	0	0	0	0	1	1	0	0	0	0	1	0	0	0
ND10	0	0	1	1	0	0	1	0	0	0	0	1	1	1	0	0	0	1	0	0	0
ND11	1	0	0	1	1	0	0	0	0	0	1	0	0	1	0	0	1	1	1	0	1
ND12	1	0	1	1	1	0	0	0	0	0	1	0	0	1	0	0	1	1	1	0	1
ND13	1	1	0	0	1	0	0	0	1	1	0	1	1	1	0	0	1	1	1	1	1
ND14	1	1	1	0	1	0	0	0	1	1	0	1	1	1	0	0	1	1	1	1	1
ND15	0	1	0	0	0	0	1	0	1	0	0	1	1	1	0	0	0	1	0	1	0
ND16	0	1	1	0	0	0	1	0	1	0	0	1	1	1	0	0	0	1	0	1	0
ND17	1	1	0	0	1	0	0	0	1	0	1	0	0	1	0	0	1	1	1	1	1
ND18	1	1	1	0	1	0	0	0	1	0	1	0	0	1	0	0	1	1	1	1	1
ND19	0	1	0	0	0	0	0	0	1	0	0	1	1	1	0	0	0	1	0	1	0
ND20	0	1	1	0	0	0	0	0	1	0	0	1	1	1	0	0	0	1	0	1	0